10 EXAM PRACTICE

1 A halogenoalkane has the structure $CH_3CHClCH(CH_3)_2$.

Which response shows its name and classification?

A 2-chloro-3-methylbutane primary

B 2-chloro-3-methylbutane secondary

C 2-methyl-3-chlorobutane secondary

D 2-methyl-3-chlorobutane tertiary [1]

(Total for Question 1 = 1 mark)

2 These are the equations for two hydrolysis reactions of halogenoalkanes:

reaction 1 $CH_3CH_2CH_2Br + H_2O \rightarrow CH_3CH_2CH_2OH + HBr$

reaction 2 $CH_3CH_2CH_2Cl + H_2O \rightarrow CH_3CH_2CH_2OH + HCl$

Which statement explains why reaction 1 is faster than reaction 2, when both reactions occur under the same conditions?

A Bromine has a higher atomic radius than chlorine.

B Chlorine is more electronegative than bromine.

C The carbon-chlorine bond has a greater bond enthalpy than the carbon-bromine bond.

D The relative atomic mass of bromine is greater than that of chlorine. [1]

(Total for Question 2 = 1 mark)

3 A mixture of 1-bromobutane, potassium cyanide and ethanol is heated under reflux to prepare a sample of a nitrile.

Which formula shows the structure of the organic product in this preparation?

A $CH_3CH_2CH_2CN$ **C** $CH_3CH(CN)CH_3$

B $CH_3CH_2CH_2CH_2CN$ **D** $CH_3CH_2CH(CN)CH_3$ [1]

(Total for Question 3 = 1 mark)

4 Alcohols can be converted into chloroalkanes in different ways.

Which reagent is the most suitable for this conversion?

$CH_3CH_2CH_2CH_2OH \rightarrow CH_3CH_2CH_2CH_2Cl$

A Cl_2 **C** PCl_5

B HCl **D** H_3PO_4 [1]

(Total for Question 4 = 1 mark)

5 Alcohols can be oxidised to aldehydes, ketones or carboxylic acids.

Which practical method is the most suitable to use in this oxidation?

$CH_3CH_2CH_2CH_2OH \rightarrow CH_3CH_2CH_2COOH$

A distillation with addition **C** heating under reflux

B fractional distillation **D** simple distillation [1]

(Total for Question 5 = 1 mark)

6 An organic compound with the formula $CH_3CH_2CH_2CH_2Cl$ is heated under reflux with NaOH(aq) for some time.

What is the formula of the main organic product of this reaction?

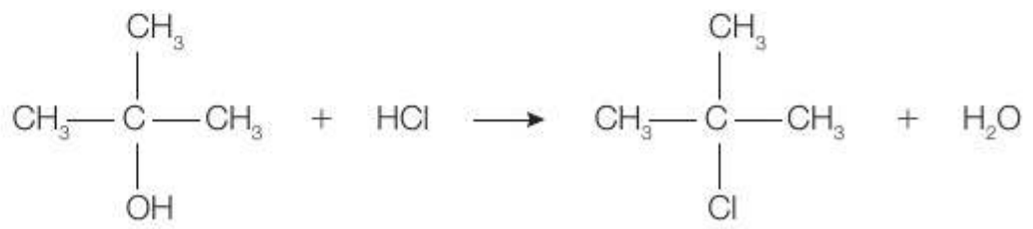
A $CH_3CH_2CH_2OH$ **C** $CH_3CH{=}CH_2$

B $CH_3CH_2CH_2CH_2OH$ **D** $CH_3CH_2CH{=}CH_2$ [1]

(Total for Question 6 = 1 mark)

7 The following is a description of how to prepare a dry, pure sample of 2-chloro-2-methylpropane starting from methylpropan-2-ol.

The equation for the reaction is:



methylpropan-2-ol 2-methyl-2-chloropropane

Step 1: Place about 9 cm^3 of methylpropan-2-ol into a separating funnel and carefully add 20 cm^3 of concentrated hydrochloric acid, about 3 cm^3 at a time. After each addition, hold the stopper tap firmly in place, and invert the funnel a few times. Then, with the funnel in the upright position, loosen the stopper briefly to release any pressure.

Step 2: Leave the separating funnel and contents in a fume cupboard for about twenty minutes. Shake gently at intervals.

Step 3: Allow the layers in the separating funnel to separate and then run off, and discard, the lower aqueous layer.

Step 4: Add sodium hydrogencarbonate solution 2 cm^3 at a time. Shake the funnel carefully after each addition and release the pressure of gas by loosening the stopper. Repeat until no more gas is evolved.

Step 5: Allow the layers to separate and then run off, and discard, the lower aqueous layer.

Step 6: Run the organic layer into a small, dry conical flask and add some anhydrous sodium sulfate. Swirl the flask occasionally for about five minutes.

Step 7: Carefully decant the organic liquid from the solid sodium sulfate into a pear-shaped flask. Add a few anti-bumping granules and set up the flask for distillation. Collect the liquid that distils over between 47–53 °C.

(a) Other than wearing safety spectacles and a protective coat, explain one safety precaution you should take when using concentrated hydrochloric acid. [2]

(b) State the purpose of adding sodium hydrogencarbonate in step 4. [1]

(c) State the purpose of adding anhydrous sodium sulfate in step 6. [1]

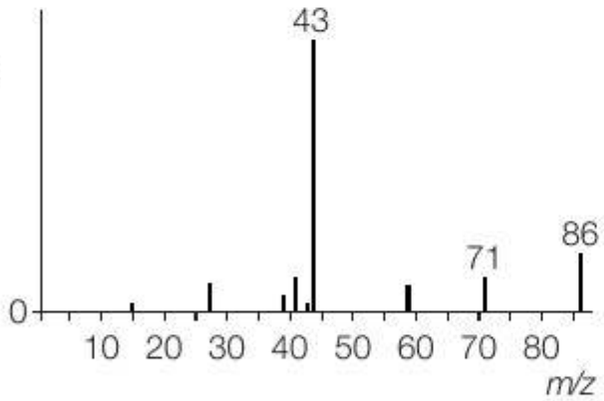
(d) Draw a labelled diagram of the distillation of the organic liquid in step 7. [4]

(e) A student started with 7.40 g of methylpropan-2-ol and obtained 7.82 g of 2-chloro-2-methylpropane. Calculate the percentage yield of the product. [3]

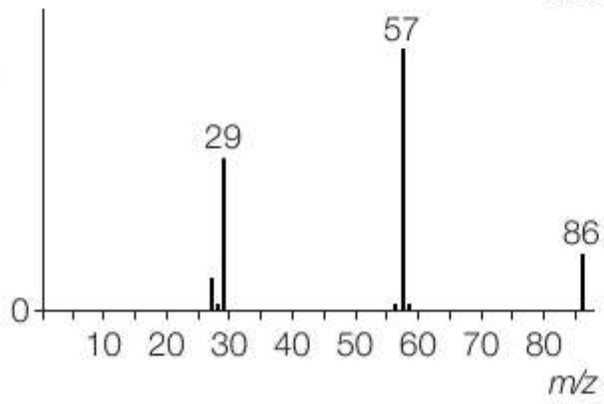
(Total for Question 7 = 11 marks)

8 Spectrum A and Spectrum B are the mass spectra of pentan-2-one ($CH_3COCH_2CH_2CH_3$) and pentan-3-one, ($CH_3CH_2COCH_2CH_3$), but not necessarily in that order.

A



B



Explain which spectrum belongs to each compound. [4]

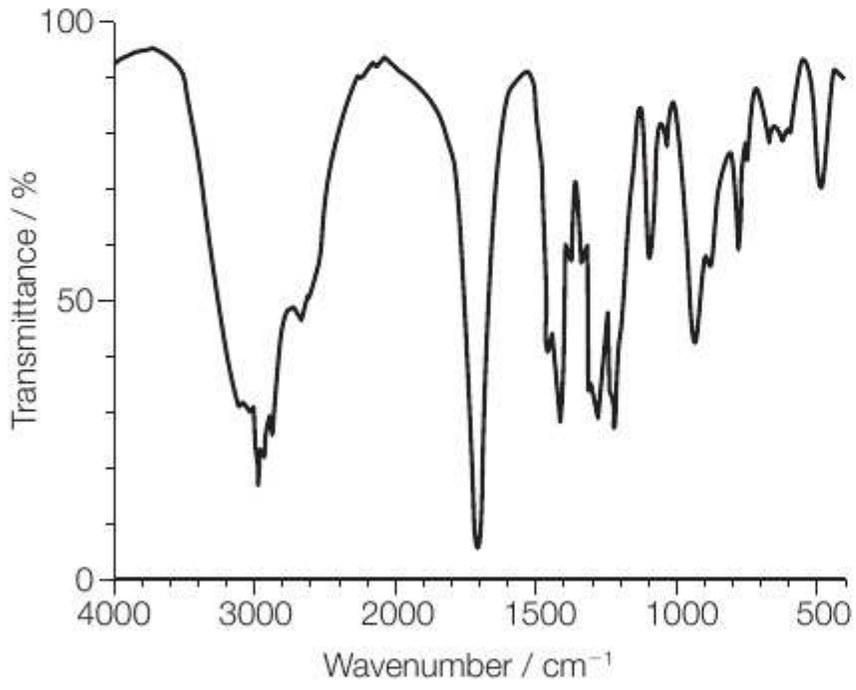
(Total for Question 8 = 4 marks)

9 A primary alcohol can be oxidised by reaction with acidified potassium dichromate(VI). The major product obtained depends on the conditions used.

If the oxidising agent is slowly added to the alcohol and then the product is distilled off as it forms, an aldehyde is collected.

If the alcohol is heated under reflux with an excess of the oxidising agent, a carboxylic acid is formed.

The infrared spectrum below is that of a product formed by the oxidation of butan-1-ol.



(a) Identify the product and explain your reasoning. [3]

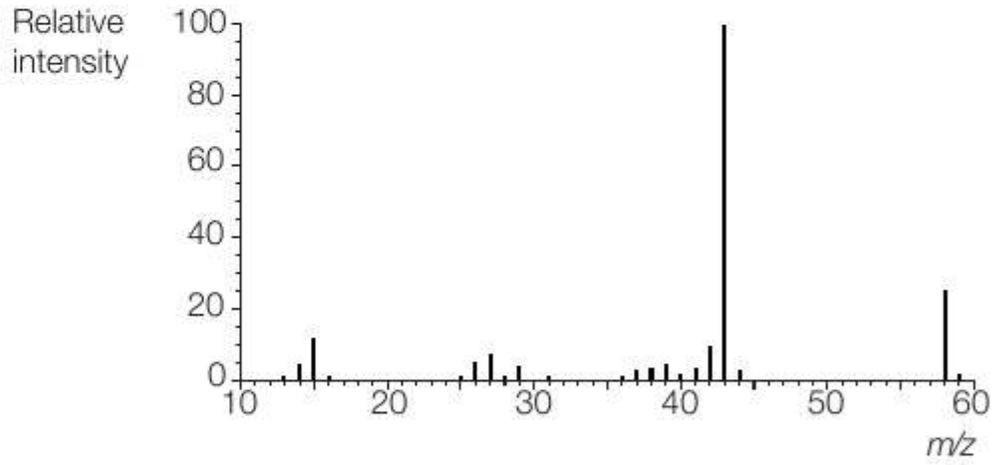
(b) Write an equation for the oxidation of butan-1-ol to this product. Use [O] to represent the oxidising agent. [2]

(Total for Question 9 = 5 marks)

10 Compound X has the following composition by mass: C, 62.07%; H, 10.34%; O, 27.59%.

(a) Calculate the empirical formula of compound X. [3]

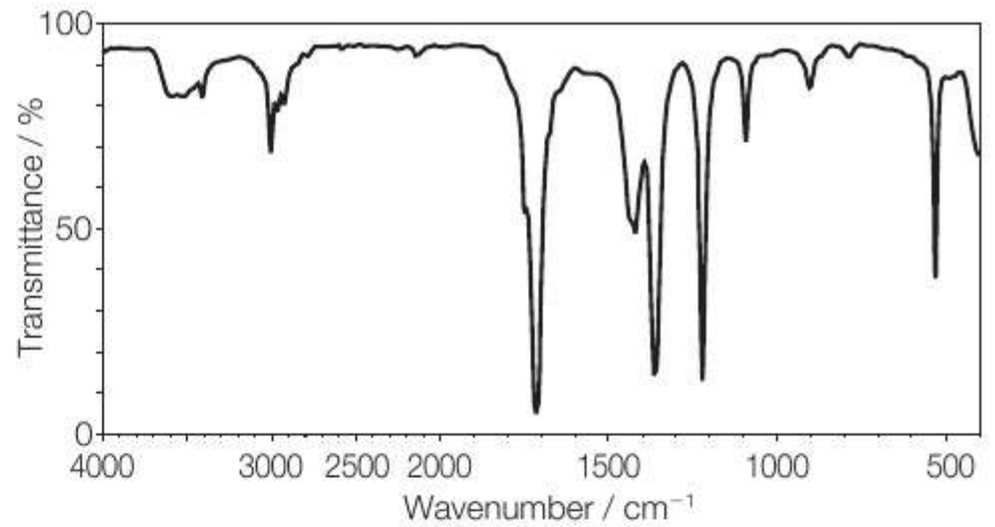
(b) The mass spectrum of compound X is shown below.



Deduce the molecular formula of compound X. Show how you obtained your answer. [3]

(c) Compound X is one of two structural isomers. One is an aldehyde, the other is a ketone.

The infrared spectrum of compound X is shown below.



Compound X produced no observable change when heated with Fehling's solution.

(i) Identify compound X. [1]

(ii) Identify the species responsible for the peaks at 15 and 43 in the mass spectrum of compound X. [2]

(iii) Identify the functional group responsible for the absorbance at 1700 cm^{-1} in the infrared spectrum of compound X. [1]

(Total for Question 10 = 10 marks)